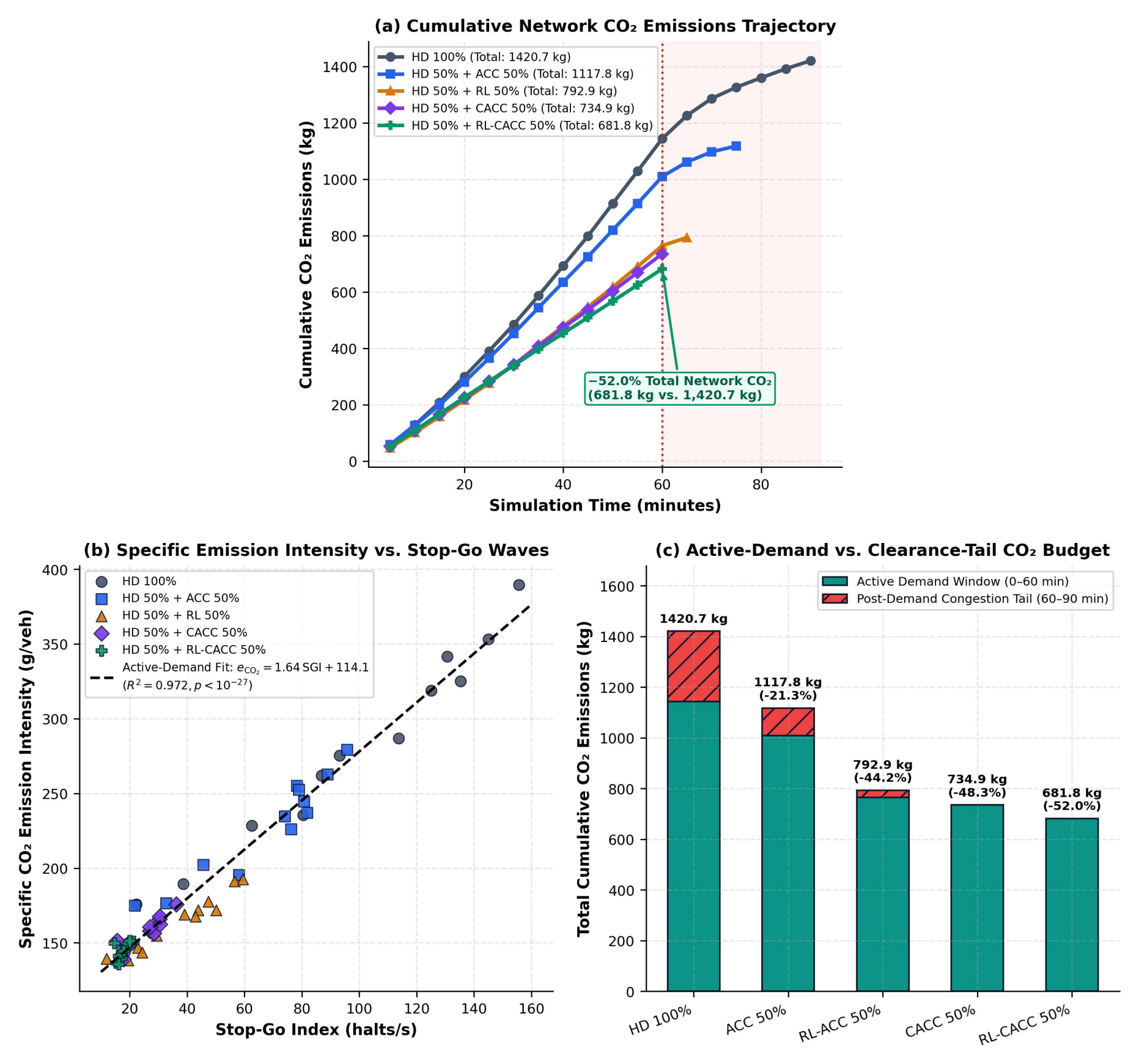

(a) Cumulative Network CO₂ Emissions Trajectory
HD 100% (Total: 1420.7 kg)
HD 50% + ACC 50% (Total: 1117.8 kg)
HD 50% + RL 50% (Total: 792.9 kg)
HD 50% + CACC 50% (Total: 734.9 kg)
HD 50% + RL-CACC 50% (Total: 681.8 kg)
−52.0% Total Network CO₂
(681.8 kg vs. 1,420.7 kg)
Cumulative CO₂ Emissions (kg)
Simulation Time (minutes)
(b) Specific Emission Intensity vs. Stop-Go Waves
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
Active-Demand Fit: $e_{CO_2} = 1.64\,SGI + 114.1$
$(R^2 = 0.972, p < 10^{-27})$
Specific CO₂ Emission Intensity (g/veh)
Stop-Go Index (halts/s)
(c) Active-Demand vs. Clearance-Tail CO₂ Budget
Active Demand Window (0–60 min)
Post-Demand Congestion Tail (60–90 min)
1420.7 kg
1117.8 kg
(-21.3%)
792.9 kg
(-44.2%)
734.9 kg
(-48.3%)
681.8 kg
(-52.0%)
Total Cumulative CO₂ Emissions (kg)
HD 100%
ACC 50%
RL-ACC 50%
CACC 50%
RL-CACC 50%